Runbook RB-112: Deploying the ledger service

Owner	Payments platform team
Escalation	Ledger on-call engineer, then the platform duty manager
Last reviewed	2 October 2026
Expected duration	40 minutes, of which 25 minutes are waiting for the rollout

1. Purpose

This runbook describes how to release a new version of the ledger service to production. It applies to scheduled releases only; emergency fixes follow runbook RB-090.

2. Prerequisites

- An approved change ticket with a release window of at least one hour.
- Access to the production cluster through the bastion host:
  - a valid hardware key registered to your account;
  - membership of the ledger-deployers group.
- A second engineer available to watch the dashboards during the rollout.

3. Procedure

3.1 Freeze the release

Check out the release tag and confirm that it was signed by the release manager.

```
$ git fetch --tags origin
$ git checkout release/2026.10.2
$ ./scripts/verify-signature.sh release/2026.10.2
```

3.2 Apply the configuration

Review the values below against the change ticket before applying them. The pool size must never exceed the connection limit agreed with the database team.

```
ledger:
  replicas: 4
  database:
    pool_size: 32
    statement_timeout_ms: 5000
  feature_flags:
    async_settlement: true
```

3.3 Roll out

1. Announce the start of the release in the operations channel.
2. Restart the deployment and follow its progress:
   a. pods are replaced one at a time;
   b. each new pod must pass its readiness check before the next one starts.
3. Stop and go to section 4 if any pod fails to become ready within five minutes.

```
kubectl -n ledger rollout restart deployment/ledger-api
kubectl -n ledger rollout status deployment/ledger-api --timeout=10m
```

3.4 Verify

Run the query below fifteen minutes after the last pod has started. Every settlement created in that period should be in the SETTLED or PENDING state.

```
SELECT status, count(*)
  FROM settlements
 WHERE created_at > now() - interval '15 minutes'
 GROUP BY status;
```

> **After the rollout**
> - Close the change ticket and record the release tag.
> - Post the verification query output in the ticket.
> - Hand the dashboards back to the on-call engineer.

4. Rollback

Roll back without waiting for approval when any threshold in the table is crossed.

Symptom	Threshold	Action
Error rate on /settle	above 2% for 5 minutes	kubectl rollout undo, then page the duty manager
Settlement latency (p99)	above 1,500 ms	Scale to 6 replicas; roll back if no change in 10 minutes
Database connections	above 90% of the limit	Reduce pool_size to 24 and redeploy
Pods not ready	any pod after 5 minutes	Roll back immediately

5. Contacts

Role	*Team*	*Hours*
Ledger on-call engineer	Payments platform	24/7
Database duty engineer	Data infrastructure	07:00 to 22:00
Platform duty manager	Platform engineering	24/7

6. Change history

Rev.	Date	Change
7	2 Oct 2026	Added the database connection threshold
6	14 Jun 2026	Replaced the manual canary with a rolling restart
5	3 Feb 2026	Moved the verification query to the read replica